Shangyang Min

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EDUCATION

Brown University | Master of Science (Sc.M.) in Computer Science **09/2023 – 05/2025**

- GPA: 4.0/4.0
- Artificial Intelligence/Machine Learning
- Specialization: Multimodal, Brain-Inspired AI, Cross-domain Applications

Michigan State University | B.S. in Computer Science Engineering with high Honor **08/2019 – 05/2023**

- GPA:3.95/4.0
- Specialization: Software Development, Machine Learning, Computer Vision, Biomedical Imaging

SKILLS

- **Research Areas**: Deep Learning, Biomedical Engineering
- **Keywords**: CNNs, Transformers, Diffusion, Multimodal, Representation Learning, Radiomics
- **Additional Interests & Proficiencies**: LLM, HCI, Game Dev, Computational Neuroscience

Research

Lee Lab | Research Assistant **09/2024 – 05/2025**

- Investigate Brain-Computer-Interface integration with gaming, created a real-time EEG and Unity pipeline that enables natural avatar control.
- Explored multimodal fusion techniques combining vision and EEG models for improved interaction fidelity.

Human Augmentation and Artificial Intelligence Laboratory | Research Assistant **05/2022 – 08/2023**

- Developed Feature Imitating Networks for biomedical imaging task, improving model robustness to limited annotations.
- Led the research project while mentoring two undergraduate students.

Henry Ford Health System | Research Assistant **09/2022 – 09/2023**

- Collaborated on a NIH-funded research program between Henry Ford Health System and Michigan State University.
- Conducted machine learning analysis on tumor detection and radiomics features from DCE-MRI scans.

PROJECTS

Game Development Studio **09/2021 – 05/2023**

- Designed and Developed game mechanics and AI behaviors for multiple game projects
- Worked under mentorship from Iron Galaxy Studio professionals, gaining professional development workflow experience.

Selected Research Projects

- **Trigger-Simulation Backdoor Removal Defense**: Proposed virtual-prompt realignment that slashed attack success rate from 93% to 0% backdoored Llama-2 variants within 7 epochs while keeping MMLU 42% accuracy.
- **VQA with Action Insights**: Integrated external SOTA vision models and tokenized video-info annotations into GPT-4/Vicuna prompts, achieved high performance with zero-shot GPT-4 vs. fine-tuned SeViLa.
- **See full list in my project webpage**. Recent submissions and projects, topics including multimodal, LLM, computer vision, computational neuroscience and HCI prototypes, etc.

PUBLICATIONS

- **S. Min**, H. B. Ebadian, T. Alhanai and M. M. Ghassemi, *Feature Imitating Networks Enhance the Performance, Reliability and Speed of Deep Learning on Biomedical Image Processing Tasks*, *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Orlando, FL, USA, 2024, pp. 1-5, doi: 10.1109/EMBC53108.2024.10782373.